

APPLIED AI & ROBOTICS · HOUSTON CITY COLLEGE

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Building Intelligent Systems 4 From NLP Pipelines to Neural Networks

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From Foundations to NLP

Six courses building toward a complete understanding of intelligent systems 4 from AI philosophy to deep learning and natural language processing.



FEATURED PROJECTS

NewsBot³ NewsBot 2.0

An NLP pipeline that evolved from a midterm prototype into a full-featured, multilingual final project.

NewsBot 4 NLP Midterm

Text classification with Naive Bayes and TF-IDF.
Sentiment analysis, POS tagging, syntax parsing, and
named entity recognition.

- 96.3% accuracy on BBC News dataset
- NLTK, spaCy, scikit-learn, pandas

NewsBot 2.0 4 NLP Final

Upgraded end-to-end NLP pipeline. Adds LDA topic
modeling, LSA summarization, semantic search,
multilingual translation, and a conversational chatbot
interface.

- 96%+ accuracy · 1,490 articles · 4 languages
- NLTK, spaCy, scikit-learn, sumy, langdetect, deep-translator

96%+ Accuracy

4 Languages

English · German · French · Spanish

[Project Repository](#)

[Live Demo](#)

Technical Skills

- 1

Python
- 2

NLTK
- 3

spaCy
- 4

scikit-learn
- 5

PyTorch
- 6

Hugging Face
- 7

VS Code
- 8

Google Colab
- 9

HTML/CSS
- 10

CNNs
- 11

RNNs
- 12

Transformers
- 13

Multi-agent Systems
- 14

NLP Pipelines
- 15

AI Cybersecurity
- 16

Database Design

What Makes Me Unique



Multilingual by Nature

Born and raised in Germany, fluent in multiple languages 4 built multilingual AI because I understand what language means personally.



Design Meets AI

Trained in graphic design and 3D design at a vocational college in Germany 4 I build systems that work and look professional.



Security Aware

Studied AI Cybersecurity 4 I understand the risks behind the systems I build, not just how to build them.

Looking Back, Moving Forward

What I Learned

- How to build real AI systems from scratch
- The full ML workflow end-to-end
- How chatbots, translators, and agents work under the hood
- The ethics and security side of AI

Where I'm Headed

- Build a web app for NewsBot 2.0
- Explore BERT and transformer models deeper
- Apply AI skills professionally
- Keep growing this portfolio